UCUG1903 Signals and Information Technology

2026-27 Fall Term, Assignment 2

In this assignment, you will use MATLAB to generate audio signals from sinusoids. Building on Assignment 1, you will synthesize a short melody from numbered musical notation (简谱), using only the fundamental frequency for each note. Please **upload a single ZIP file** containing your MATLAB scripts, the generated wav file, and a Word or PDF report to Canvas by September 16, 2026, at 23:59. Late submissions are accepted but will incur a 50% deduction per day.

Task: Sinusoid to Sound (Melody Synthesis) [100]

a) Problem description

A pure tone can be modeled as a sinusoidal signal. In music, each note corresponds to a specific fundamental frequency (pitch). Numbered musical notation (简谱) uses the degrees 1 through 7 to represent the notes of a major scale. The mapping between these degrees, their corresponding note names, and their frequencies is given in Table 1 below:

note	Do	Re	Mi	Fa	So	La	Ti
degree	1	2	3	4	5	6	7
Freq(Hz)	261.63	293.66	329.63	349.23	392.00	440.00	494.00

Each tone can be generated as a sinusoid at its corresponding frequency. Therefore, you can synthesize a melody by combining these sinusoidal signals.

In this problem, you are provided with four melodies to choose from. You will select one, generate its audio, and listen to the resulting song. Each note is assigned a beat count, with one beat assumed to last 1 second. A dash ("-") in the score indicates a *hold*, meaning the duration of the preceding note is extended by one beat. Use the information in Table 1 to synthesize your chosen melody.

Select one of the following scores:

Song A: 1 1 5 5 | 6 6 5 - | 4 4 3 3 | 2 2 1 -

Song B: 1 2 3 1 | 1 2 3 1 | 3 4 5 - | 3 4 5 -

Song C: 1 2 3 1 | 5 - - - | 6 6 1 6 | 5 - - -

Song D: 5 3 5 3 | 5 3 1 | 2 4 3 2 | 5 -

b) Implementation instructions

Open the files in the task1 folder. Function interfaces (inputs and outputs) are already defined. Complete the TODO steps marked by error ('...:notImplemented').

Step 1: Generate one pure tone

Open generate_tone.m and complete the function to generate a single tone at a pure frequency. The function receives a note degree as input and should return a sinusoid of the specified duration at the corresponding frequency.

Hints:

- Use the generate_sinusoid function you created in Assignment 1. You can either copy the generate_sinusoid .m to the current folder or add the corresponding path.
- Create a lookup table to map note degrees (1–7) to their corresponding frequencies.

Step 2: Build the full melody

Open synthesize_melody.m and complete the function that constructs the entire melody using the generate_tone function you created in Step 1.

Hints:

- To build the melody, generate each tone individually and concatenate them into a single audio vector.
- For note like 5 -, you should generate a single note with longer duration.
- In MATLAB, a melody is simply a larger array – you can either pre-allocate it and fill the correct segments later, or directly concatenate the generated signals.
- A for loop is an efficient way to iterate through the note sequence.

The provided output-generation code will generate an audio file at output/task1_pure_sine_melody.wav and plot a portion of the waveform. **In your submission, include the generated WAV file. In your report, include a screenshot of the plot, state which song you chose, and briefly describe what you hear.**

for : **for loop to repeat specified number of times**

Syntax

```
for index = values
  statements
end
```